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Studierichting: Informatica
Oefeningenreeks 1D nr. 21.5

$A(z) = z^3 + z + 10$

$A(-2) = -8 - 2 + 10 = 0 \Rightarrow z_1 = -2$

$$
\begin{array}{r|rrr|r}
 & 1 & 0 & 1 & 10 \\
-2 & \downarrow & -2 & 4 & -10 \\
\hline
 & 1 & -2 & 5 & 0
\end{array}
$$

$A(z) = (z+2)(z^2 - 2z + 5) \rightarrow D = (-2)^2 - 4 \cdot 5 = -16$

$z_{2,3} = \dfrac{2 \pm 4i}{2} \Rightarrow z_2 = 1 + 2i, z_3 = 1 - 2i$

$z_{1,2,3} \in \{-2, 1 + 2i, 1 - 2i\}$

References